North Central London (NCL) ICS

Asthma Primary Care Clinical Pathway in Adults (for patients aged ≥18)

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Approving Body	NCL ICB CAG
NCL-Wide / Borough(s)	NCL-Wide
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Key

- Pi — "Must do" actions for GP's / (Triaged by RSS)
- B — Recommendations for Primary Care
- R — Red flag / urgent referral
- O — Routine referral
- P — Public health intervention
- G — Audio-visual aids for patients and GP
- Click icon for clinical evidence
- Right-click to use hyperlink

Note: Alpha-numeric step references are to aid printing in black & white and colour blindness

Investigations must be within 6 months of date of referral

1.0 (B)

Initial assessment of patient presenting with suspected asthma
Structured clinical assessment (from history and examination of previous medical records) look for:
- Recurrent episodes; symptoms of wheeze, cough, breathlessness and chest tightness
- Symptom variability or seasonal variation in symptoms
- History of smoking tobacco and other substances (note secondary tobacco smoking exposure)
- History of occupational (symptoms better when days away from work or on holiday) - refer
- Organic triggers
- Recorded observation of wheeze
- Personal or family history of atopic disease
- Historical record of variable PEF of FEV
- Code suspected asthma if applicable and update after objective tests

3.0 (B)

Objective investigations:

1) Spirometry. A FEV1/FVC ratio <0.7 AND z score (see Appendix 3) for the ratio is < -1.65 demonstrates obstructive airway disease. NOTE: spirometry may be affected by empirical treatment with inhaled corticosteroids

2) FeNO (if available). A level >40ppb is regarded as positive. NOTE: A normal FeNO does not exclude asthma. Smoking and inhaled corticosteroid use can reduce FeNO. For further information [**1**]

3) Bronchodilator reversibility (BDR) test. An improvement in FEV1 of 12% or more, together with an increase in volume of 200 ml or more, is regarded as a positive test. An improvement of >400ml in FEV1 is strongly suggestive of asthma

4) Peak flow variability of 20% over two weeks

2.0 (R)

Red flag symptoms 2WW:
- Unexplained haemoptysis (age ≥ 40 years)
- chest x-ray suggesting lung cancer or mesothelioma

4.0 (B)

Code asthma as per QOF requires a record of spirometry plus one other objective test: FeNO or reversibility or variability:
- a FeNO level of 40 ppb or more with either positive BDR or positive peak flow variability or
- positive BDR and positive peak flow variability irrespective of FeNO level

Code asthma with fixed airflow obstruction if positive BDR but FEV1/FVC <0.7 AND z score (see Appendix 3) for the ratio is < -1.65 post bronchodilator and if never smoked tobacco.

If significant tobacco smoking history consider **code asthma and COPD** (clinical judgement).
COPD pathway coming soon

5.0 (B)

Diagnostic uncertainty
Variable peak expiratory flow (PEF) readings can support an asthma diagnosis if there is diagnostic uncertainty after initial assessment, a FeNO test, and/or objective tests to detect airway obstruction. A value of more than 20% variability after monitoring at least twice daily for 2-4 weeks is regarded as a positive result.

Positive PEF → 4.0 (B)

Positive PEF → 6.0 (B)

Negative PEF → 7.0 (G)

Negative PEF → 8.0 (O)

6.0 (B)

Code suspected asthma if symptoms suggestive of asthma, obstructive spirometry and:
- negative BDR and FeNO (if available) level between 25 and 39 ppb and positive peak flow variability, or
- positive BDR, a FeNO (if available) level between 25 and 39 ppb and negative peak flow variability.

Review the diagnosis after 6 to 10 weeks by repeating spirometry and objective measures of asthma control and reviewing symptoms.

7.0 (G)

Consider seeking Advice & Guidance prior to referral

8.0 (O)

Review the diagnosis after 6 to 10 weeks by repeating spirometry and objective measures of asthma control and reviewing symptoms.
Investigate for other likely diagnosis spirometry & **refer as appropriate (or if clinical uncertainty)**
Consider referral to Royal Brompton in the event of suspected occupational asthma

9.0 (B)

Consider watchful waiting if asymptomatic or commencement of treatment with assessment of response (particularly if airway obstruction present)

10.0 (G)

Check patient demonstrates good inhaler technique with chosen inhaler device. Training videos available on Asthma UK [**2**]

11.0 (G)

Provide self-management education and a personalised asthma action plan [**3**]

eductional videos re inhaler technique are available [**4**]

12.0 (B)

Management
- Pharmacotherapy as per appendix 1 & 2. Start at a level most appropriate to initial severity. Review any treatment initiated at 4-8 weeks. Consider referral to community pharmacy for inhaler technique check and/or new medicines service to support compliance and inhaler technique
- Where a dry powder inhaler (DPI) is not suitable and a metered dose inhaler (MDI) is chosen consider use of spacer device
- Discuss features of poor control and check patient understands their treatment
- Minimise numbers/types of inhaler devices and ensure prescribing is by brand. See appendix 2 for preferred brands in line with the green agenda.
- Offer annual flu vaccine and pneumonia vaccine (where appropriate) and ensure up-to-date covid vacination

13.0 (G)

Lifestyle advice popup

Smoking status = smoker - (**Ask Advise Act - http://elearning.ncsct.co.uk/vba-stage_1**)
Alcohol consumption = <14 units/per week
Physical inactivity / diet, aim for BMI <30, <150 mins per week
Mental wellbeing = Signs of stress / anxiety

13.1 (P)

14.0 (B)

Assess control in 4 - 8 weeks

15.0 (B)

Good Control
- No daytime symptoms
- No night-time awakening due to asthma
- No need for rescue medication
- No asthma attacks
- No limitations on activity including exercise
- Normal lung function (in practical terms FEV1 and/or PEFR > 80% predicted or best
- Minimal side effects from medication

16.0 (B)

Poor control
- Daytime symptoms ≥ 3 times a week
- Night-time awakening ≥ 1 per week
- The use of reliever medication ≥ 3 times per week
- Asthma attacks ≥ 1 per year (course oral prednisolone or emergency attendance)

17.0 (B)

Review inhaler technique, adherence and collection of prescriptions. Consider if a diagnosis of asthma is correct (e.g. COPD with asthma), whether the patient is still being exposed to irritants and confirm smoking status/ passive smoke exposure

Poor control → 18.0 (B)

Poor control → 19.0 (B)

18.0 (B)

Continue with current therapy and conduct annual review

Good control → 18.0 (B)

19.0 (B)

Maintain control by increasing treatment as necessary (see appendix 1 & 2). Review control 4-8 weeks after any change in therapy. Use lowest effective doses to achieve control.

Persistent poor control → 21.0 (B)

20.0 (B)

Annual review (**utilise asthma EMIS template**)
- Assess symptoms using the Asthma Control Test [**5**] or Asthma Control Questionnaire [**6**] and frequency of reliever use
- Assess lung function e.g. PEF rate and smoking status
- Document frequency and severity of any asthma attacks and time off work as a result of asthma
- Confirm number of hospital admissions, number of SABA inhalers , oral steroid and antibiotic courses issued in last year
- Check adherence and inhaler technique and demonstrate good technique.
- Confirm possession of/use of personalised action plan
- For those on long term (e.g. >3 months) steroid tablets or requiring frequent courses (e.g. 3-4 courses/year) – See NICE CKS [**7**]

Good control → 22.0 (B)

21.0 (B)

Persistent poor control:
- Despite high dose ICS/LABA
- ≥ 3 - 6 SABA inhalers in the last 12 months despite primary care review
- More than 2 asthma attacks requiring oral steroids in the last 12 months
- Life-threatening asthma attack
- Consideration of oral macrolide treatment as prophylaxis

22.0 (B)

- Continue with current treatment **OR**
- Consider step down of treatment if patient is well controlled for 2-3 months
- See NCL adult asthma inhaler choice interim guidance for further information on stepping down treatment

Consider seeking Advice & Guidance prior to referral → 23.0 (G)

24.0 (O)

Refer to specialist

26.0

Key:
BDR: bronchodilator reversibility
FeNO: fractional exhaled nitric oxide
ICS: Inhaled corticosteroid
LABA: long acting β agonist
LAMA: long acting muscarinic antagonists
LTRA: leukotriene receptor antagonist
SABA: short acting β agonist
SR: slow release

25.0

Pregnancy
Please refer to: BTS/SIGN guideline: Asthma in pregnancy, chapter 12, SIGN 158, July 2019 [**8**]

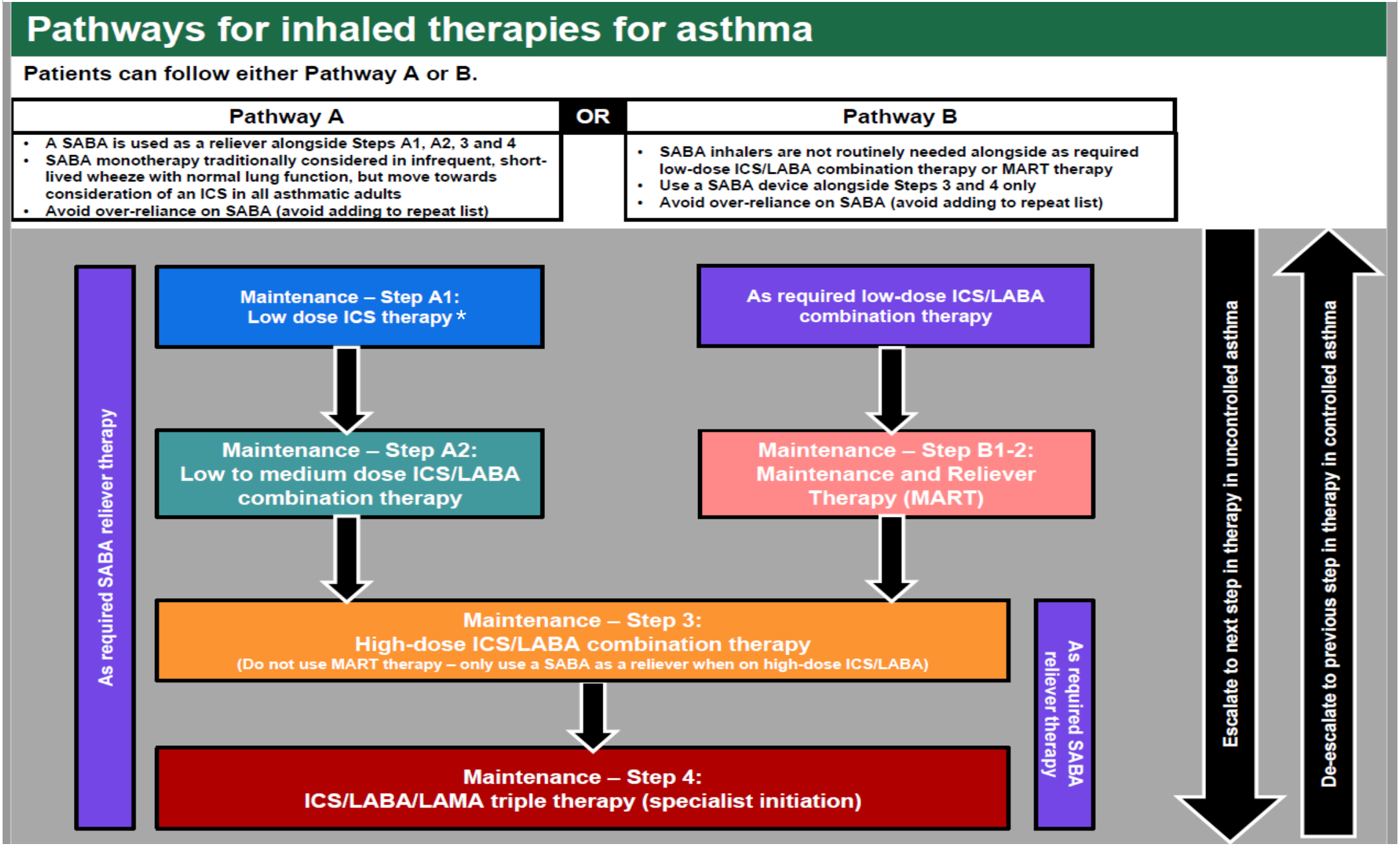
27.0

Acute asthma
Please refer to: BTS/SIGN guideline: Management of acute severe asthma in adults in general practice [**8**] – algorithm page 164, Annex 3, SIGN 158, July 2019

28.0

[**1**]: https://www.e-lfh.org.uk/programmes/feno-in-asthma/
[**2**]: https://www.asthma.org.uk/advice/inhaler-videos/
[**3**]: https://www.asthma.org.uk/advice/manage-your-asthma/action-plan/
[**4**]: https://www.asthma.org.uk/advice/inhaler-videos/
[**5**]: https://www.asthma.com/understanding-asthma/severe-asthma/asthma-control-test/
[**6**]: https://www.qoltech.co.uk/acq.html
[**7**] https://cks.nice.org.uk/topics/corticosteroids-oral/#!scenario
[**8**]: https://www.brit-thoracic.org.uk/quality-improvement/guidelines/asthma/

Appendix 1: Adult asthma treatment algorithm - see full guidance for further information [9]



*If asthma remains uncontrolled on low dose ICS, an LTRA (montelukast 10mg OD) may sometimes be used if there are exercise induced or rhinitis symptoms (note: montelukast may cause neuropsychiatric reactions or sleep disturbance, so extra caution should be taken in patients with history of related conditions); it should only be continued if deriving benefit

Appendix 2: Adult asthma inhaler choices (see full guidance for further information [9]) –https://www.ncl-mon.nhs.uk/wp-content/uploads/3_Asthma_inhaler_choice.pdf

Inhalers on the NCL Joint Formulary

PATHWAY A

Maintenance Step 1: Low-dose ICS

Pulmicort Turbohaler
Budesonide
100mcg device: 2 doses BD
200mcg device: 1 dose BD
CO_2: LOW — DPI

Flixotide Accuhaler
Fluticasone propionate
50mcg device: 2 puffs BD
CO_2: LOW — DPI

Clenil Modulite
Beclomethasone
100mcg device: 2 doses BD
CO_2: HIGH — pMDI

Flixotide Evohaler
Fluticasone propionate
50mcg device: 2 puffs BD
CO_2: HIGH — pMDI

MaintenanceStep 2: Low to medium dose ICS/LABA

Fostair Nexthaler
Beclometasone/ Formoterol
100/6 mcg device
Low dose: 1 dose BD
Med dose: 2 doses BD
CO_2: LOW — DPI

Relvar Ellipta
Fluticasone furoate/ Vilanterol
92/22 mcg device
No low dose
Med dose: 1 dose OD
CO_2: LOW — DPI

Symbicort Turbohaler
Budesonide/ Formoterol
200/6 mcg device
Low dose: 1 dose BD
Med dose: 2 doses BD
Other devices available
CO_2: LOW — DPI

Fostair pMDI
Beclometasone/ Formoterol
100/6 mcg device
Low dose: 1 dose BD
Med dose: 2 doses BD
CO_2: HIGH — pMDI

Reliever therapy - SABA

Ventolin Accuhaler
Salbutamol – 200micrograms/dose
1 dose as required (max 1 dose QDS)
CO_2: LOW — DPI

Easyhaler Salbutamol
Salbutamol – 6 month expiry once opened
1-2 puff(s) as required
CO_2: LOW — DPI

Salamol CFC-Free inhaler
Salbutamol – contains ethanol
1-2 puff(s) as required
CO_2: HIGH — pMDI

Salamol Easi-Breathe
Salbutamol
1-2 dose(s) as required
CO_2: HIGH — BAI

Next steps for PATHWAYS A or B

Step 3: High dose ICS/LABA

Fostair Nexthaler
Beclometasone/ Formoterol
200/6 mcg device: 2 doses BD
CO_2: LOW — DPI

Relvar Ellipta
Fluticasone furoate/ Vilanterol
184/22 mcg device: 1 dose OD
CO_2: LOW — DPI

Symbicort Turbohaler
Budesonide/ Formoterol
400/12 mcg device: 2 doses BD
CO_2: LOW — DPI

Fostair pMDI
Beclometasone/ Formoterol
200/6 mcg device: 2 doses BD
CO_2: HIGH — pMDI

Step 4: Triple therapy (ICS/LABA/LAMA)

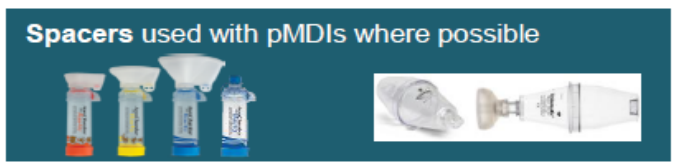
Enerzair Breezhaler
Indacaterol/ Glycopyrronium/ Mometasone
114/46/136 mcg device: 1 dose OD
CO_2: LOW — DPI

High-dose ICS/LABA with Spiriva Respimat (Tiotropium)
High-dose ICS/LABA plus Spiriva Respimat 2.5 mcg device: 2 doses OD
CO_2: LOW — SMI

Trimbow
Beclometasone/ Formoterol/ Glycopyrronium
87/5/9 mcg device: 2 puffs BD
172/5/9 mcg device: 2 puffs BD
CO_2: HIGH — pMDI

Additional reliever therapy – SABA – choices as per Pathway A

Always prescribe by brand only



PATHWAY B

Reliever therapy: As required low-dose ICS/LABA combination therapy (additional SABA not required)

Fostair Nexthaler
Beclometasone/ Formoterol
100/6 mcg device: 1 dose PRN (up to 8 doses daily)
CO_2: LOW — DPI

Symbicort Turbohaler
Budesonide/ Formoterol
100/6mcg or 200/6mcg device: 1 dose PRN (up to 8 doses daily)
CO_2: LOW — DPI

Fostair pMDI
Beclometasone/ Formoterol
100/6 mcg device: 1 dose PRN (up to 8 doses daily)
CO_2: HIGH — pMDI

Maintenance Step 1-2: Maintenance and Reliever Therapy

Fostair Nexthaler
Beclometasone/ Formoterol
100/6 mcg device: 1 dose BD and 1 dose PRN (up to 8 doses daily)
CO_2: LOW — DPI

Symbicort Turbohaler
Budesonide/ Formoterol
100/6mcg device: 1 dose BD & 1 dose PRN
200/6mcg device: 1 dose BD & 1 dose PRN
Usually up to 8 doses daily; max 12 doses daily
CO_2: LOW — DPI

Fostair pMDI
Beclometasone/ Formoterol
100/6 mcg device: 1 dose BD and 1 dose PRN (up to 8 doses daily
CO_2: HIGH — pMDI

29.0

Inhaler choice:
- Device choice should be guided by the patient's ability to use the device and other potential issues that may affect compliance e.g. spacer use/once daily vs twice daily dosing. Refer to NICE decision inhaler decision aid [**10**]
- Combination inhalers are recommended where available
- Prescribing by brand is recommended to ensure consistency of device supplied
- Aim for device consistency across therapy to enhance patient compliance
- How to use your inhaler | Asthma UK [**2**]
- Before initiating a new drug therapy practitioners should check adherence with existing therapies, check inhaler technique, and eliminate trigger factors

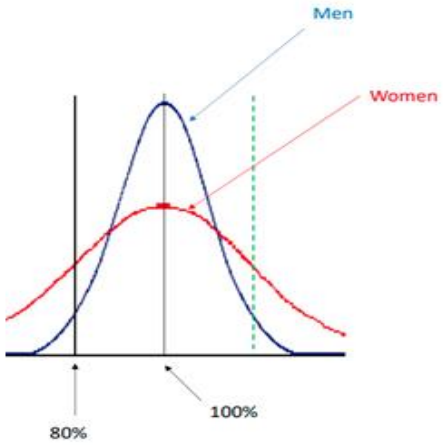
Stepping down ICS:
- Consider stepping down treatment if good control for 2-3 months; Reduce ICS by 25-50% every 3 months
- Stepping down should be explained to the patient and the patient's personalised asthma action plan updated accordingly
- After treatment is reduced, review the patient within 4-8 weeks

[**9**]: https://www.ncl-mon.nhs.uk/wp-content/uploads/3_Asthma_inhaler_choice.pdf
[**10**]: https://www.nice.org.uk/guidance/ng80/resources/inhalers-for-asthma-patient-decision-aid-pdf-6727144573#:~:text=The%20National%20Institute%20for%20Health,how%20to%20improve%20their%20technique.

Appendix 3: Spirometry and Z scores

What are the limitations of using % predicted?

Height, sex, age and ethnicity are all factors involved in the determination of predicted lung volume. Using the same cut of value e.g. 80% predicted assumes the variation amongst healthy populations is the same between men and women which is incorrect.



- The blue data (men) have the same mean but there is less variation than the red data (women)
- 80% predicted for men should be considered worse (less of the population included) than 80% predicted for women as the spread of the data is greater

Why are Z scores necessary?

An appropriate interpretation of the spirometric data requires the use of a population-specific reference equation. However, the most widely used equations were established in European populations with limited age groups. The extrapolation of these equations, based on a specific population, and their uses for a different population lead to measurement and interpretation biases. The benefit of Z scores is that it is completely independent of age, height, and sex

What are Z scores and how are they calculated?

The Z score (also called standardised residual (SR)) is calculated by the ratio of the difference between the measured value and that predicted with the residual standard deviation. It describes how many standard deviations a subject is deviated from its reference value.

Z score = (measured – predicted)/standard deviation

90% of the population is included in 1.645 standard deviations above and below the mean (average). It misses out the top and bottom 5% which are deemed to be abnormally high or low results. A negative SR means the result is below predicted. A SR value of 0 means the result is identical to the predicted and a SR of −1.645 puts the result on the fifth percentile and is at the lower limit of normal (LNN)

How are z scores interpreted?

Severity classification in airflow obstruction is a two stage process:

- The FEV1/FVC must be below the LLN (Z score < -1.645) to be classified as obstructive.
- Severity grading is then based on FEV1 Z-score as per the table below

Threshold for Z score	Severity grading	Approx. chance of finding this result in healthy population
<−1.645	Mild	1 in 20
<−2	Moderate	1 in 40
<−2.5	Moderately Severe	1 in 150
<−3	Severe	1 in 750
<−4	Very Severe	1 in 30,000

Example:

		Pred	Act 1	%Pred	SR
KCOc_SB	mmol(min*kPa*L)	1.44	1.10	76.5	-1.06

In the above example, despite the predicted value being 76.5%, the SR value is greater than -1.645 (the bottom 5% of healthy population) and therefore is normal.

Reference

Sylvester KP, Clayton N, Cliff I et al 2020. ARTP statement on pulmonary function testing 2020. BMJ Open Resp Res, Volume 7:e000575

Appendix 4: Asthma spirometry and FeNO commissioned services

NOTE: Many of the respiratory services below are either pilots or new hublets and subject to change. Please access the GP website for up-to-date information on how to access and refer to these sites. The GP website will be updated with service details in due course.

Borough	Site	Status of service provision	Provision of spirometry or FeNO or both	Area covered	How to refer patients
All NCL boroughs	Finchley Memorial CDC (interim hublet)	LIVE	Spirometry and FeNO	All NCL boroughs	See GP website for full details
Barnet	Finchley Memorial CDC (interim hublet)	LIVE	Spirometry and FeNO	All NCL boroughs	See GP website for full details
Camden	Bloomsbury Surgery hublet South Camden & Islington hublet – Site TBC	LIVE Pilot due to be launched soon	Spirometry and FeNO Spirometry and FeNO	Central Camden PCN only South Camden PCNs only	Referral form (embedded into emis) completed and emailed to: ccpcn.mdt-asthma@nhs.net TBC
Enfield	Carlton House Evergreen Surgery	LIVE Going LIVE imminently	Spirometry Spirometry & FeNO	Enfield Unity and Enfield Care PCN with an aim for Pan Enfield service provision	Practice will email PCN team with a referral form. This is then risk assessed and will move to cross organisational booking. Contact for service: renata.chavda@nhs.net
Haringey	Currently no local services available. Please refer patients to the NCL commissioned Finchley Memorial CDC hublet Please note: Whittington Health Community Services are only commissioned to provide spirometry tests for COPD patients only				
Islington	North Islington hublet - Andover Medical Centre and Goodinge Group Practice South Camden & Islingtc hublet – Site TBC	Pilot due to be launched soon Pilot due to be launched soon	Spirometry and FeNO Spirometry and FeNO	North Islington PCNs only South Islington PCNs only	Practices will book into the clinic directly. Contacts for service: r.habtom1@nhs.net TBC. Contact for service: nsage@nhs.net